GENOMELENS

Economic analysis of a genomic report

Cost–utility model of acting on the findings in one whole-genome input, valued at $100,000 per QALY from a healthcare sector perspective.

Source input	**public benchmark genome huDEDF57**	Input type	**WGS**
Generated	**24 October 2026**	Time horizon	**each decision model states its own**
Discounting	**3.0%, costs and QALYs**		

Proprietary.
Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.

Public benchmark genome.
A public benchmark genome. It is a real person's genome; the findings are shown to demonstrate the analysis and are not a clinical report.

Research and educational use — not medical or financial advice.
This is a decision-analytic model, not a clinical or economic evaluation submitted to any authority. Its parameters are population averages applied to one genome: it values the expected consequences of acting on a pattern of findings, and does not predict what will happen to any individual. Monetary figures include health valued at a willingness-to-pay threshold and are not cash returned to anyone. Clinically relevant findings require confirmation by an appropriate professional — a physician, clinical pharmacist or board-certified genetic counselor — before anything follows from them.

Economic impact overview

Modeled economic impact

NO HEADLINE VALUE IN THIS GENOME

No primary modeled value to headline: this genome's modeled values are shown below with their class, because they are measured nulls. Every other finding is classed with its reason in the table of all economically relevant findings.

2	2	3	2
TESTING-STRATEGY / POPULATION-LEVEL VALUE	TRUE ZERO	UNQUANTIFIED	TECHNICALLY UNRESOLVED

Modeled values (shown separately, never added)

TRUE ZERO EVIDENCE-QUALIFIED · TRUE ZERO

$0

OPRM1 finding → OPRM1-guided naltrexone. A measured null: the evidence finds no significant outcome difference

SUPPORTED CALCULATION AT THE ASSUMED AGE

Modeled net monetary benefit at assumed age 45: **$0** EVIDENCE-QUALIFIED · TRUE ZERO — the supported calculation at the assumed age: only decisions whose every model input is sourced (here a measured null, which does not depend on the age: the supported decisions show no measured effect on health, and no cost of acting is charged, which is not an unknown). A stricter test, not a total of the values above, and it does not replace them.

Research and educational use only; not medical, clinical or financial advice. Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results.

All economically relevant findings

Every economic result of this genome, ranked within its class. Values in different classes are never added.

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
UGT1A1 finding	UGT1A1-guided irinotecan dosing	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
Hypothetical, based on assumptions: any adult, before results	Population screening for Lynch syndrome, hereditary breast and ovarian cancer and familial hypercholesterolemia	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
OPRM1 finding	OPRM1-guided naltrexone	**$0**	TRUE ZERO **EVIDENCE-QUALIFIED · TRUE ZERO**	A measured null: the evidence finds no significant outcome difference
vitamin D (at age 45)		**valued at zero**	TRUE ZERO	Outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older …
ACADVL pathogenic (ClinVar)		**not quantified**	UNQUANTIFIED	No economic model yet
UGT1A1 Poor Metabolizer (PM, *28/*28 — Gilbert syndrome phenotype) (irinotecan)		**not quantified**	UNQUANTIFIED	Actionable finding, person-level value not quantified: DPWG starts a UGT1A1 *28/*28 (poor metabolizer) patient at a …
Nonsyndromic Cleft Lip / Palate carrier (uncertain)		**not quantified**	UNQUANTIFIED	No disease model of its own yet
CYP2C9 Intermediate Metabolizer (IM) (warfarin)		**not assessable**	TECHNICALLY UNRESOLVED	Warfarin dosing needs both CYP2C9 and VKORC1 called
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)		**not assessable**	TECHNICALLY UNRESOLVED	CYP2D6 copy number not assessed: a SNV/indel callset cannot see the *5 deletion or gene duplications …

All economically relevant findings (continued)

The evidence hierarchy

- **Evidence-supported**: every model input is sourced and the value does not rest on an age that is stated rather than observed (badge EVIDENCE-QUALIFIED)
- **Primary modeled**: the selected economic model for a decision; it may depend on the stated assumptions shown beside it
- **Once the decision arises**: the value per person if the decision arises; not an expected value today
- **Alternative modeled**: a competing model or sensitivity scenario for the same decision; never added to the primary result
- **Conditional**: valid only if a stated clinical or treatment condition holds; never added
- **Family / cascade-testing**: accrues to relatives or another beneficiary; never in the individual's personal total
- **Population / testing strategy**: a value per person screened or tested in a population, not this donor's value; never added **in this genome**
- **Hypothetical / projected**: an exploratory scenario on stated assumptions, not an observed or established benefit; in no total **in this genome**
- **Negative**: modeled, and the costs exceed the health gain valued at $100,000 per QALY; shown, never hidden
- **True zero**: the evidence supports no incremental benefit or change in care within the defined decision **in this genome**
- **Unquantified**: potentially relevant, but the evidence is insufficient for a monetary value; not $0 **in this genome**
- **Technically unresolved**: the genomic evidence cannot reliably support the interpretation the decision needs; not $0 **in this genome**
- **Not applicable**: the model or intervention does not apply to the assessed context (sex, age or setting); not $0

Unknown is not $0. Unquantified is not $0. Technically unresolved is not $0. No supported estimate is not a measured zero. Negative is not zero. An alternative is never added.

Findings valued by their decision models

Net monetary benefit (NMB) is the QALYs gained valued at $100,000 each, less the change in healthcare cost, in 2019 US dollars. It is a health-economic value, not cash saved: a positive NMB can come with higher spending. Each clinical decision is valued once; the categories are never added to each other. Each figure's horizon is its own model's (the carrier models run over a lifetime; no pooled disease model ran for this genome).

Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Age 45 is stated for this run, not observed (one demonstration age for every sample; the PGP public profiles give only an age bracket, and the GIAB and 1000 Genomes samples publish no age). A value whose model depends on age is modeled at that age; a value its source gives for one age only is held at that age, and its row says so. Model evidence is stated apart from whether its context matches the donor, and the qualification stays assumption-driven; an age-dependent value would be evidence-qualified only at an observed or self-reported age.

Sex: known male.

Supported calculation at assumed age 45

$0 EVIDENCE-QUALIFIED · TRUE ZERO · AGE 45 ASSUMED every decision supported at the assumed age is a measured null, so the supported calculation is a measured zero. Measured nulls eligible under the available donor context, each with its age restriction, are listed in A. Whether it applies to this donor is not established: the donor's age is not observed.

- no decision is supported at the assumed age beyond its measured nulls

A. Evidence-qualified expected value

$0 EVIDENCE-QUALIFIED · TRUE ZERO every evidence-qualified result is a measured null (a trial found no effect) or no change in care, so this is a measured zero, not an absence; the modeled values are in B.

- **OPRM1-guided naltrexone**: $0 EVIDENCE-QUALIFIED · TRUE ZERO
  measured null in alcohol use disorder: genotype x treatment P = .32, so no health gain and no cost of acting is credited.
  comparator: naltrexone regardless of genotype; sources: PMID 25760804.

B. Other modeled expected value, with further limitations

- none

C. Values that hold only in a stated clinical context

- none for this genome

D. Alternatives and scenarios, shown separately

- none for this genome

E. Actionable pathways not yet quantified

- ACADVL pathogenic (ClinVar) — actionable economically unquantified: no economic model yet.
- UGT1A1 Poor Metabolizer (PM, *28/*28 — Gilbert syndrome phenotype) (irinotecan) — actionable economically unquantified: DPWG starts a UGT1A1 *28/*28 (poor metabolizer) patient at a reduced irinotecan starting dose, but the economic evaluation (Butzke 2016) is an average per patient tested whose strategy may also reduce heterozygotes' doses (for whom DPWG needs no action), with no result for *28/*28, so no value is estimated for this genotype (shown only as a testing-strategy value).
- Nonsyndromic Cleft Lip / Palate carrier (uncertain) — evidence insufficient: no disease model of its own yet.
- CYP2C9 Intermediate Metabolizer (IM) (warfarin) — technically unresolved: warfarin dosing needs both CYP2C9 and VKORC1 called; VKORC1 is not called in this input, so the warfarin decision is not priced.
- CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants) — technically unresolved: CYP2D6 copy number not assessed: a SNV/indel callset cannot see the *5 deletion or gene duplications (the CYP2D7 pseudogene region), so this call is technically unresolved; only a poor-metabolizer call from two no-function alleles, which neither changes, is kept.

Findings valued by their decision models (continued)

F. Valued at zero: care does not change (a result, not a gap)

- **vitamin D (at age 45)** — valued at zero: outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older (LeBoff 2022, PMID 35939577); Endocrine Society 2024 (Demay, PMID 38828931): for adults younger than 50 it suggests against empiric vitamin D beyond the dietary reference intake and against routine 25(OH)D testing

Findings-first economics

No pooled result: no finding enters the pooled model, so there is no combined estimate to state; each finding is valued by its own decision model, by category, on the next page.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

HEALTH GAIN

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

HEALTHCARE COST CHANGE

Not available

no pooled result

MODELED NET BENEFIT

Not available

no pooled result

DECISION UNCERTAINTY

Not available

no pooled result

0

findings carry a standardized economic estimate

16 of the 16 findings shown carry no standardized estimate, and none is assigned one the model cannot support. Separately, 1 clinical decision(s) are valued by their own decision models (1 evidence-qualified, 0 modeled with stated assumptions), by category in the section 'Findings valued by their decision models'; a finding with such a value has its figure there, not here.

OPRM1 OPRM1 A118G (mu-opioid… | Vitamin D Synthesis Efficiency | Nonsyndromic Cleft Lip / Palate… | UGT1A1 Poor Metabolizer (PM… | CYP2D6 Intermediate Metabolizer | CYP2C9 Intermediate Metabolizer | OPRM1 — Opioid-response… | CHRNA5 | CRHR1 — Stress-axis variant | CFTR carrier | MT-ND5 pathogenic | SPG7 pathogenic | SCAMP4 pathogenic | PRSS1 pathogenic | CAVIN1 pathogenic | EDA pathogenic

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype finding with a priced economic pathway; a finding whose decision the evidence values at zero is listed with the true zeros.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk and prevention findings, in detail

No risk or prevention finding of this genome has a standardized estimate, so there is no card to show. The findings are reported without one below, each with its reason on the glance pages; a lower evidence grade changes how a finding should be read, not whether it is shown.

Reported without a standardized value (14): Vitamin D Synthesis Efficiency..., Nonsyndromic Cleft Lip / Palate carrier..., UGT1A1 Poor Metabolizer (PM, *28/*28..., CYP2D6 Intermediate Metabolizer (IM)..., CYP2C9 Intermediate Metabolizer (IM)..., CHRNA5 — Nicotine-dependence..., and 8 more. Each carries its reason on the glance pages (no change in care, not assessable, unquantified, or reproductive information); no value is invented.

How the findings combine

Taken together, what do these findings mean?

There is no pooled reference case for this genome: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. Findings bearing on one condition are combined here before any value is credited; with nothing in the pool there is nothing to combine, and no figure stands in for the absent result. The findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'.

INCREMENTAL HEALTHCARE COST

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

INCREMENTAL QALYS

Not available

no pooled result

REFERENCE-CASE NMB

Not available

no pooled result

CLASSIFICATION

Not available

no pooled result

How sure is this, and what rests on judgment?

Uncertainty has two layers: the spread of modeled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across simulations

No probabilistic analysis of the pooled model: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. No probability of cost-effectiveness, interval or one-way swing is stated for it, because there is no pooled strategy to sample. The decision models carry their own parameter uncertainty (2,000 draws from the ranges the sources print, rescaled to a per-carrier value where a source prints another unit); where a decision has a range, its 5th to 95th percentile is printed beside that decision's value.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Base case: the value the published evidence supports for this genome, or “not estimable” with the reason. Scenario: what GenomeLens computes on its own assumptions, graded by the quality of its inputs, with each assumption numbered in the log below. Scenario figures are not from published data for this genome and are not the evidence headline; the reproductive costs belong to a prospective child and are in no personal total. A row valued by its own decision model repeats that model's value as a pointer; the model's basis (published, or hypothetical) is stated beside the headline. This genome's assessment is incomplete (a finding is actionable but unquantified, technically unresolved, or lacks the evidence to value): that is not a conclusion that the genome has no further economic value.

Table 1. Results by finding

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
This genome, to the person sequenced (each finding once)	**Not established** Incomplete assessment: not zero	**None projected** Sum of the rows not marked 'not summed': model and scenario values, never this donor's total	**no evidence-supported value; no value from decision models that are not evidence-qualified; no projected finding is summed (a placeholder on its own row is not added); figures marked 'not summed', reproductive values and payer values are excluded**
OPRM1 OPRM1 A118G (mu-opioid receptor) — OPRM1 A118G — opioid/naltrexone response	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Valued once, by the decision model(s) of this report: 'OPRM1-guided naltrexone' (a measured null) Sources reviewed: PMID 22784013; PMID 33387367; PMID 31961981
Vitamin D Synthesis Efficiency — Nutrient-metabolism variant — deficiency prevention	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Care does not change under a guideline (outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older ... Sources reviewed: PMID 17634462; PMID 32533132; PMID 35727271
Nonsyndromic Cleft Lip / Palate carrier (uncertain)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: no value: the evidence does not support one: no disease model of its own yet
UGT1A1 Poor Metabolizer (PM, *28/*28 — Gilbert syndrome phenotype) (irinotecan)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: actionable, with no economic estimate yet: actionable finding, person-level value not quantified: DPWG starts a UGT1A1 ...
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)	**Not estimable** Not estimable	**Not estimable** Conditional on a genotype call this input cannot establish; no figure is shown	No earlier figure; the decision models' verdict: not assessable from this input: CYP2D6 copy number not assessed: a SNV/indel callset cannot see the *5 deletion or gene ...
CYP2C9 Intermediate Metabolizer (IM) (warfarin)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: not assessable from this input: warfarin dosing needs both CYP2C9 and VKORC1 called; VKORC1 is not called in this input ...
OPRM1 — Opioid-response pharmacogenomic variant	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Valued once, by the decision model(s) of this report: 'OPRM1-guided naltrexone' (a measured null)

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
CHRNA5 — Nicotine-dependence susceptibility variant	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care
CRHR1 — Stress-axis variant (CRHR1/FKBP5)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care
CFTR carrier — heterozygote health	**Not estimable** Not estimable	**No model yet** No model yet	Heterozygote health effects are real but not costed per gene
MT-ND5 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
SPG7 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
SCAMP4 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
PRSS1 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
CAVIN1 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
EDA pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
IRF6 Cleft lip / palate Nonsyndromic Cleft Lip / Palate; undeclared; carrier	**1 in 2** that a child inherits it, per pregnancy	**Not available** no cost of the child's condition is registered for IRF6; only the test that resolves the risk is priced	**—** per pregnancy	**$185** a known-variant test of the pregnancy or the child	GenomeLens registered estimate

Methods

How these figures were produced, and how to read them

A decision-analytic model values the expected consequences of acting on each finding in this genome. The figures are model outputs under stated assumptions, kept apart from clinical outcomes.

ANALYTIC FRAME

Perspective	**healthcare sector**
Comparator	**each decision model states its own**
Outcomes	**cost, QALYs, NMB**
Time horizon	**each decision model states its own**
Discounting	**3%, costs and QALYs**
Willingness to pay	**$100,000/QALY**
Currency	**USD, United States**
Uncertainty	**decision models: 2,000-draw parameter uncertainty; pooled model: no PSA (no pooled model)**

HOW TO READ EACH FIGURE

- **Evidence-supported value:** every model input is sourced and the value does not rest on an age that is stated rather than observed. Where the age is stated, as for every public genome, an age-dependent value is a standardized scenario and is labeled so.
- **Decision-model rows:** computed from published inputs with at least one assumption (for a public genome, the stated age); shown, never presented as evidence-qualified.
- **Hypothetical, based on assumptions:** what a finding could be worth if its curated assumptions held; never added to any counted total.
- **Projected, based on assumptions:** the GenomeLens scenario beside a base case that cannot be computed soundly; not the evidence headline.
- **Reproductive costs:** per pregnancy, for a prospective child, never in a personal total.

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labeled as such.
- Costs are restated to 2019 US dollars from each source's price year.
- Findings without a modeled decision pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Proprietary method. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.